

A. Datasets

cities

animals

elements

p_{true} : 'Tokyo is in Japan.'

p_{false} : 'Hanoi is in Poland.'

p_{retain} : 'Summarize the email\dots'

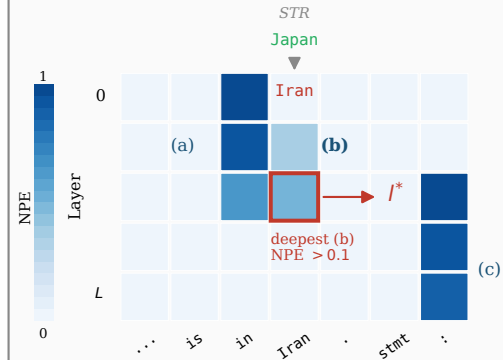
t_{true} = 'True'

t_{false} = 'False'

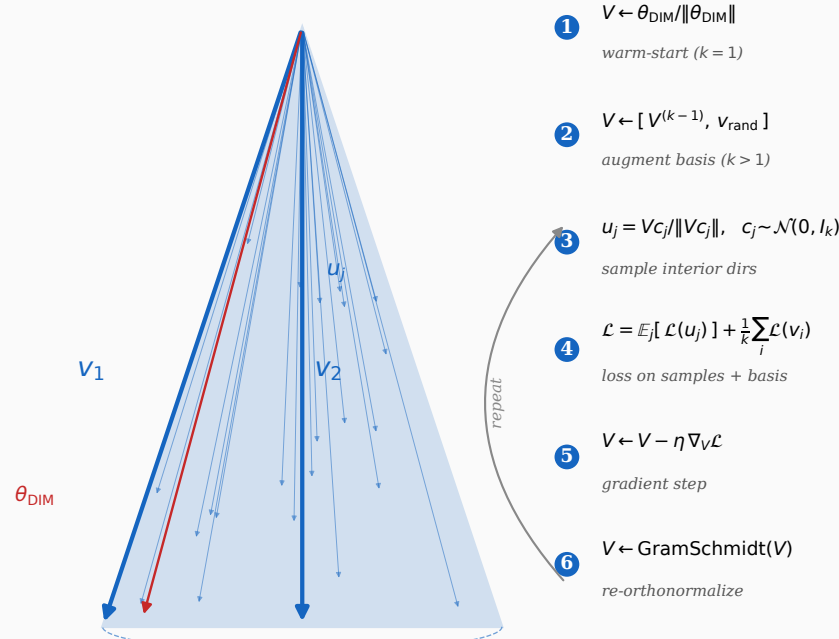
forced-choice True/False over factual triples

contrastive triples

B. Exp 1 — Localization



C. Exp 2 & 3 — TDO / TCO



TDO: $k = 1$ direction v_1 · TCO: $k \geq 2$ cone interior

frozen model · loss = $\mathcal{L}_{\text{abl}} + \mathcal{L}_{\text{add}} + \mathcal{L}_{\text{ret}}$ (necessity · sufficiency · surgicality)

D. Exp 4 — $|\cos(\theta_{\text{DIM}}, v_i)|$

v_1	0.93	0.91	0.87	0.86	0.84	0.82
v_2	0.03	0.10	0.06	0.02	0.04	0.01
v_3	0.02	0.01	0.06	0.02	0.01	0.04
v_4	0.05	0.07	0.01	0.01	0.04	0.02
	Q-1.5B	G-2B	Q-7B	G-9B	L-8B	Q-14B

bright v_1 , dark $v_{2..4}$ = novel structure

E. Exp 5 — KL retention

$k = 1$	✓	✓	×	✓	×	×
$k = 2$	✓	✓	✓	✓	✓	✓
$k = 3$	✓	✓	×	✓	×	✓
$k = 4$	✓	✓	✓	✓	✓	✓
	Q-1.5B	G-2B	Q-7B	G-9B	L-8B	Q-14B

green ≤ 0.1 (surgical), red > 0.1